Name _____

# Choosing Measures and Comparing Distributions

**Which measures should you use?** It depends on the shape of the data.

| Shape of the data | Center | Spread |
|---|---|---|
| Symmetric, no outliers | mean | standard deviation |
| Skewed, or has outliers | median | IQR |

**Why?** An outlier pulls the mean and the SD. It does not pull the median and the IQR much.

**Example:** 2, 3, 3, 4, 5, 40

| | Mean | SD | Median | IQR |
|---|---|---|---|---|
| With 40 | 9.5 | 15.0 | 3.5 | 2 |
| Without 40 | 3.4 | 1.1 | 3 | 2 |

One value moved the mean by 6.1 and the SD by about 14. The median moved 0.5. The IQR did not move.

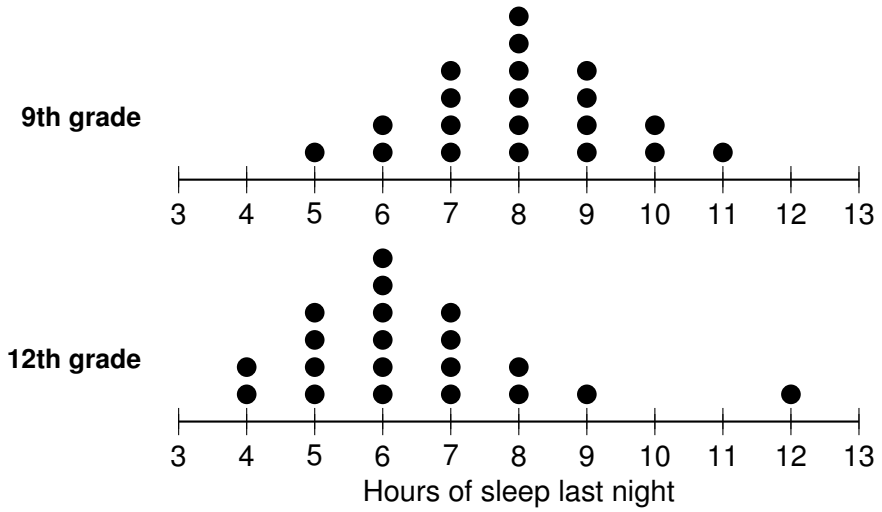
**Comparing two distributions.** Talk about all four of these. Use numbers and the context.

**Shape:** symmetric or skewed?

**Center:** which group is higher? (median or mean)

**Spread:** which group is more spread out? (IQR or SD)

**Outliers:** any values far from the rest?




| | |
|---|---|
| Shape | 9th grade is symmetric. 12th grade is skewed right. |
| Center | The median for 9th grade (8 hours) is greater than for 12th grade (6 hours). 9th graders usually sleep about 2 hours more. |
| Spread | Both groups have an IQR of 2 hours. The spreads are about the same. |
| Outliers | One 12th grader slept 12 hours. That is an outlier (the upper fence is 10). |

12th grade has an outlier, so we compare with the median and the IQR.

Work with a partner. Compare answers when you finish each problem.

1. For each data set, should you use the **mean and SD** or the **median and IQR**? Explain.

   (a) The heights of 200 adult women. The histogram is symmetric.

   (b) The price of every house sold in a city. The histogram is skewed right.

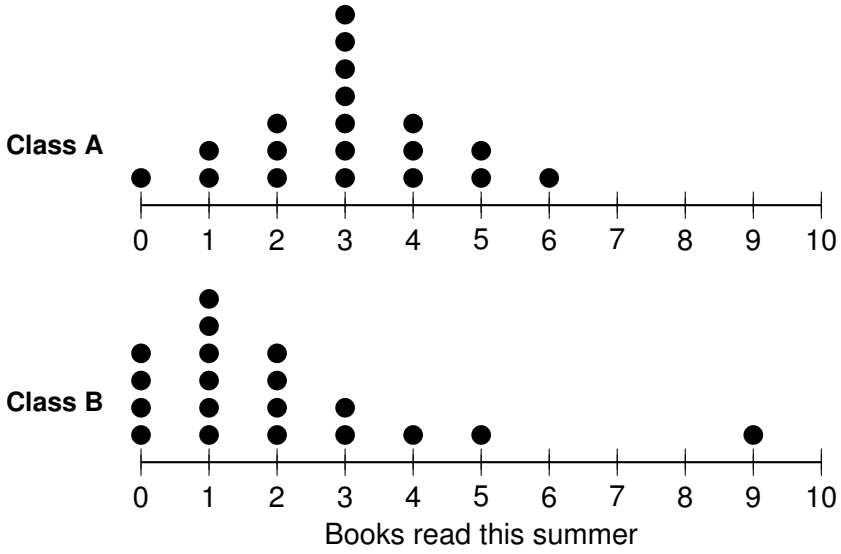
   (c) Quiz scores: 7, 8, 8, 9, 9, 10, 2

2. Look at the example on page 1: 2, 3, 3, 4, 5, 40.

   (a) Which pair describes a typical value better: mean 9.5 and SD 15.0, or median 3.5 and IQR 2? Explain.

   (b) A student says, “The SD is 15, so the values are very spread out.” Is that a fair description? Explain.

3. These dot plots show how many books students in two classes read this summer.



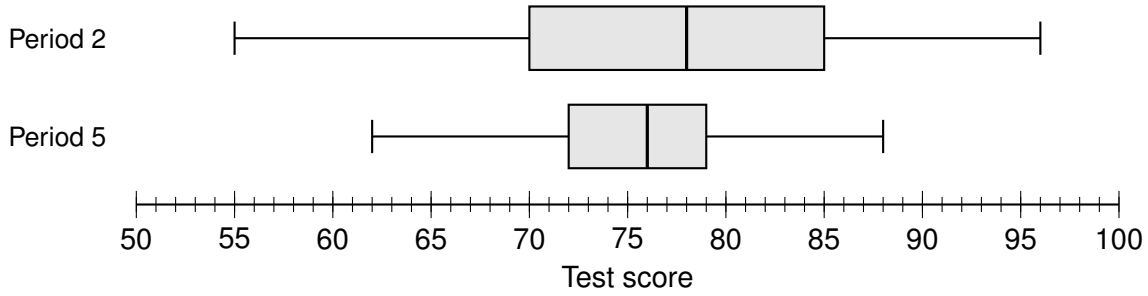

(a) Describe the shape of each class.

(b) Class A has median 3 and IQR 2. Class B has median 1 and IQR 2. Compare the centers.

(c) Compare the spreads.

(d) Does either class have an outlier? Explain.

4. These box plots show test scores in two class periods.




(a) Compare the centers. Use numbers.

(b) Compare the spreads. Use numbers.

(c) Which period's scores are more consistent? Explain.

5. Two soccer teams measured the heights of their players. Both histograms are symmetric.

| | Mean | SD |
|---|---|---|
| Team X | 70 inches | 2 inches |
| Team Y | 72 inches | 5 inches |

(a) Which team is taller on average?

(b) Which team has heights that are more spread out?

(c) Write one sentence that compares the two teams. Use both the center and the spread.